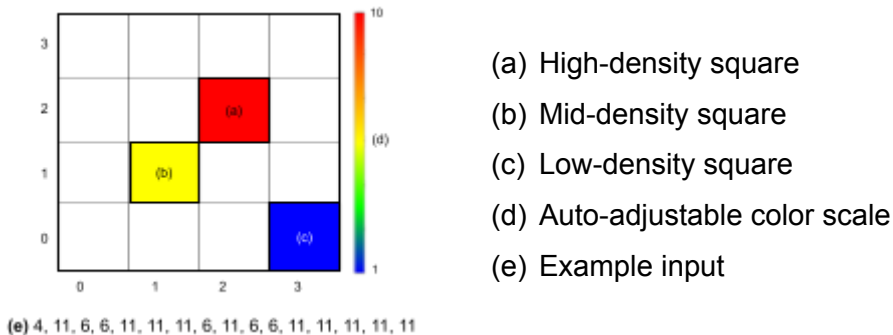


“Bin There, Done That”

Precision Neuroscience - Full-Stack Project

For this project, you’ll be asked to implement real-time visualization of streaming data. The data server will be in the cloud, with visualization on the local web client. See below for illustration of client UI:



- The client will render an NxN grid (N is configurable)
- The client will read a stream of nonnegative integers from the cloud server
- For each received number, it will be binned into the grid by zero-based remainder and quotient (ex. For 4x4 grid, 17 will go to grid cell <0,0>, 8 will go to grid cell <1,3>)
- Each grid cell will maintain a running cumulative count of hits
- A color will be assigned to each grid cell based on the running count, normalized to the maximum count (for zero count, no color will be rendered), following conventional blue-to-red heat mapping
- The grid and color scale will update in real-time as the data is streamed

You have freedom in selecting how the data generation, connection and streaming are performed.

Bonus

- How will you attempt to do this in 3D?
- Discuss the design considerations for shifting to server-side rendering